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BICYCLE COMPONENT

Abstract

A bicycle component is basically provided with a rechargeable power source and an electrical switch. The rechargeable power source is configured to be charged by electric power from an external power supply. The electrical switch is configured to be activated by only electric power from the rechargeable power source.

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Background/Summary

BACKGROUND

Technical Field

[0001] This disclosure generally relates to a bicycle component. More specifically, the present disclosure relates to a bicycle component having an electrical switch.

Background Information

[0002] In recent years, some bicycles are provided with electric components or devices to make it easier for the rider to operate the bicycle. Examples of such bicycle components include suspensions, transmission devices (e.g., derailleurs, internally geared hubs, etc.), operating devices and seatposts. Some bicycle components are provided with one or more electrical switches to operate the bicycle component. When a bicycle component includes an electrical switch, one or more electric power supplies are provided to the bicycle or the bicycle component to provide electrical power to the electrical switch.

SUMMARY

[0003] Generally, the present disclosure is directed to various features of a bicycle component having an electrical switch.

[0004] In view of the state of the known technology and in accordance with a first aspect of the present disclosure, a bicycle component is provided that basically comprises a rechargeable power source and an electrical switch. The rechargeable power source is configured to be charged by electric power from an external power supply. The electrical switch is configured to be activated by only electric power from the rechargeable power source.

[0005] With the bicycle component according to the first aspect, electric power can be reliably provided to an electrical switch of the bicycle component using a rechargeable power source that can be recharged by electric power from an external power supply.

[0006] In accordance with a second aspect of the present disclosure, the bicycle component according to the first aspect further comprises a holding structure configured to detachably hold the rechargeable power source to the bicycle component.

[0007] With the bicycle component according to the second aspect, the rechargeable power source can be removed and reinstalled.

[0008] In accordance with a third aspect of the present disclosure, a bicycle component is provided that basically comprises a rechargeable power source, an electrical switch and a holding structure. The rechargeable power source is configured to be charged by electric power from an external power supply. The electrical switch is configured to be activated by electric power from the rechargeable power source. The holding structure is configured to detachably hold the rechargeable power source to the bicycle component.

[0009] With the bicycle component according to the third aspect, electric power can be reliably provided to an electrical switch of the bicycle component using a rechargeable power source that can be removed and reinstalled.

[0010] In accordance with a fourth aspect of the present disclosure, the bicycle component according to the second aspect or the third aspect is configured so that the holding structure includes an electrical terminal structure electrically coupling the rechargeable power source to the bicycle component.

[0011] With the bicycle component according to the fourth aspect, the rechargeable power source can be easily and reliably electrically coupled to the electrical switch of the bicycle component.

[0012] In accordance with a fifth aspect of the present disclosure, the bicycle component according to any one of the first aspect to the fourth aspect is configured so that the holding structure includes an accommodating part and a power source cover. The accommodating part is configured to accommodate the rechargeable power source. The accommodating part includes an opening for removing the rechargeable power source from the holding structure. The power source cover is configured to at least partially cover the opening of the accommodating part.

[0013] With the bicycle component according to the fifth aspect, the rechargeable power source can be easily removed from the holding structure.

[0014] In accordance with a sixth aspect of the present disclosure, the bicycle component according to the fifth aspect is configured so that the holding structure includes a cover fixing structure configured to fix the power source cover relative to the accommodating part.

[0015] With the bicycle component according to the sixth aspect, the power source cover can be reliably fixed to the bicycle component for retaining the rechargeable power source to the holding structure.

[0016] In accordance with a seventh aspect of the present disclosure, the bicycle component according to any one of the first aspect to the sixth aspect further comprises a base member and an operating member. The base member is configured to be mounted to a bicycle. The operating member is movably coupled to the base member.

[0017] With the bicycle component according to the seventh aspect, a user can easily operate the electrical switch as an operating member.

[0018] In accordance with an eighth aspect of the present disclosure, the bicycle component according to the seventh aspect is configured so that the operating member is configured to operate an operated device mounted to the bicycle.

[0019] With the bicycle component according to the eighth aspect, the operating member can be used to operate an operated device mounted to the bicycle.

[0020] In accordance with a ninth aspect of the present disclosure, the bicycle component according to the seventh aspect or the eighth aspect is configured so that the operating member is configured to activate the electrical switch in response to an input to the operating member.

[0021] With the bicycle component according to the ninth aspect, the electrical switch can be easily and reliably operated in response to an input to the operating member.

[0022] In accordance with a tenth aspect of the present disclosure, the bicycle component according to any one of the seventh aspect to the ninth aspect further comprises an additional operating member movably coupled relative to the base member, the additional operating member being different from the operating member.

[0023] With the bicycle component according to the tenth aspect, the bicycle component be used to perform two distinct operations.

[0024] In accordance with an eleventh aspect of the present disclosure, the bicycle component according to the tenth aspect is configured so that the additional operating member is configured to operate an additional operated device mounted to the bicycle.

[0025] With the bicycle component according to the eleventh aspect, the bicycle component be used to operate two distinct operated devices.

[0026] In accordance with a twelfth aspect of the present disclosure, the bicycle component according to the tenth aspect or the eleventh aspect is configured so that the base member includes a proximal end configured to be mounted to the bicycle, and a distal end opposite to the proximal end. The additional operating member is pivotally supported by an axle to pivot with respect to the base member. The axle is disposed closer to the distal end than to the proximal end.

[0027] With the bicycle component according to the twelfth aspect, the additional operating member can be conveniently mounted to the bicycle for easy operation.

[0028] In accordance with a thirteenth aspect of the present disclosure, the bicycle component according to any one of the tenth aspect to the twelfth aspect is configured so that the electrical switch is provided to at least one of the operating member and the additional operating member.

[0029] With the bicycle component according to the thirteenth aspect, the bicycle component can be relatively compact by providing the electrical switch to at least one of the operating member and the additional operating member.

[0030] In accordance with a fourteenth aspect of the present disclosure, the bicycle component according to any one of the tenth aspect to the thirteenth aspect further comprises a hydraulic unit

configured to operate the additional operated device in response to an input to the additional operating member.

[0031] With the bicycle component according to the fourteenth aspect, the bicycle component can be used for operating a hydraulically operated device in response to an input to the additional operating member.

[0032] In accordance with a fifteenth aspect of the present disclosure, the bicycle component according to any one of the tenth aspect to the fourteenth aspect is configured so that the operating member is movably mounted to the additional operating member.

[0033] With the bicycle component according to the fifteenth aspect, a user can easily operate the operating member and the additional operating member.

[0034] In accordance with a sixteenth aspect of the present disclosure, the bicycle component according to any one of the first aspect to the fifteenth aspect further comprises a charging port configured to supply electrical power to the rechargeable power source from the external power supply.

[0035] With the bicycle component according to the sixteenth aspect, the rechargeable power source can be recharged without removing the rechargeable power source from the bicycle component.

[0036] In accordance with a seventeenth aspect of the present disclosure, the bicycle component according to the sixteenth aspect further comprises a cover member configured to cover the charging port in a case where the cover member is in a covering state.

[0037] With the bicycle component according to the seventeenth aspect, the charging port can be protected from contaminants when the charging port is not being used.

[0038] In accordance with an eighteenth aspect of the present disclosure, the bicycle component according to any one of the first aspect to the seventeenth aspect further comprises a wireless communicator electrically coupled to the rechargeable power source.

[0039] With the bicycle component according to the eighteenth aspect, the bicycle component can wirelessly communicate with another component or device without using a communication cable. Thus, it is possible to omit a communication cable and simplify the installation of the bicycle component.

[0040] In accordance with a nineteenth aspect of the present disclosure, the bicycle component according to the eighteenth aspect is configured so that the wireless communicator is configured to transmit information related to the rechargeable power source to an external device.

[0041] With the bicycle component according to the nineteenth aspect, a user can reliably manage the rechargeable power source.

[0042] In accordance with a twentieth aspect of the present disclosure, the bicycle component according to the eighteenth aspect or the nineteenth aspect further comprises a substrate and an antenna. The substrate supports the wireless communicator. The antenna is provided to the substrate and electrically coupled to the wireless communicator. The substrate is disposed between the rechargeable power source and the antenna.

[0043] With the bicycle component according to the twentieth aspect, the rechargeable power source and the antenna can be conventionally provided to the substrate without increasing the size of the substrate.

[0044] In accordance with a twenty-first aspect of the present disclosure, the bicycle component according to the twentieth aspect is configured so that the antenna is disposed along a peripheral edge of the substrate.

[0045] With the bicycle component according to the twenty-first aspect, reception and transmission of wireless signals can be improved.

[0046] In accordance with a twenty-second aspect of the present disclosure, the bicycle component according to the fifth aspect or the sixth aspect further comprises a wireless communicator, a substrate and an antenna. The wireless communicator is electrically coupled to the rechargeable

power source. The substrate supports the wireless communicator on a first side. The antenna is provided to a second side of the substrate and electrically coupled to the wireless communicator. The second side faces in an opposite direction to the first side. The rechargeable power source is closer to the power source cover than the antenna.

[0047] With the bicycle component according to the twenty-second aspect, the rechargeable power source can be more easily accessed for replacement.

[0048] In accordance with a twenty-third aspect of the present disclosure, the bicycle component according to any one of the twentieth aspect to the twenty-second aspect further comprises an electronic controller configured to control supply of electrical power from the rechargeable power source to the wireless communicator based on an output from a detector. The detector is configured to detect a state of at least one of the bicycle and the bicycle component.

[0049] With the bicycle component according to the twenty-third aspect, the supply of electric power from the rechargeable power source to the wireless communicator can be appropriately regulated based on an output from a detector.

[0050] In accordance with a twenty-fourth aspect of the present disclosure, the bicycle component according to any one of the first aspect to the twenty-second aspect further comprises an electronic controller configured to obtain information related to the rechargeable power source.

[0051] With the bicycle component according to the twenty-fourth aspect, information related to the rechargeable power source can be readily used by the electronic controller as needed and/or desired.

[0052] In accordance with a twenty-fifth aspect of the present disclosure, the bicycle component according to the twenty-fourth aspect further comprises a notification device electrically coupled to the electronic controller. The electronic controller is configured to control the notification device based on the information.

[0053] With the bicycle component according to the twenty-fifth aspect, a user can be easily notified about the status of the rechargeable power source.

[0054] Also, other objects, features, aspects and advantages of the disclosed bicycle component will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses preferred embodiments of the bicycle component.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0055] Referring now to the attached drawings which form a part of this original disclosure, an illustrative embodiment is shown.

[0056] FIG. 1 is a side elevational view of a bicycle that is equipped with a plurality of bicycle components in accordance with various illustrated embodiments of the present disclosure.

[0057] FIG. 2 is a front side perspective view of a handlebar area of the bicycle illustrated in FIG. 1.

[0058] FIG. 3 is a front side perspective view of an operating device and a portion of a handlebar of the bicycle illustrated in FIG. 1.

[0059] FIG. 4 is a front side perspective view, similar to FIG. 3, of the operating device mounted to the handlebar, but in which the grip cover has been removed.

[0060] FIG. 5 is a rear side perspective view of the operating device illustrated in FIG. 3.

[0061] FIG. 6 is a rear side perspective view, similar to FIG. 5, of the operating device, but in which the grip cover has been removed.

[0062] FIG. 7 is a rear end view of the operating device illustrated in FIG. 3 in which the operating members are disposed in rest positions.

[0063] FIG. 8 is a rear end view of the operating device illustrated in FIG. 7, but in which one of the operating members has been moved from the rest position of FIG. 7 to an operated position to output a first shift signal.

[0064] FIG. 9 is a rear end view of the operating device illustrated in FIGS. 7 and 8, but in which two of the operating members has been moved from their rest positions of FIG. 8 to operated positions to output a second shift signal.

[0065] FIG. 10 is an outer side elevational view of the operating device illustrated in FIG. 8, in which the operating members are disposed in their rest positions.

[0066] FIG. 11 is an outer side elevational view, similar to FIG. 10, of the operating device illustrated in FIG. 10, but in which the additional operating members has been moved from the rest position of FIG. 10 to an operated position to perform a braking operation.

[0067] FIG. 12 is a cross-sectional view of the base member of the operating device of illustrated in FIGS. 1 and 11 showing a hydraulic unit.

[0068] FIG. 13 is an outer side elevational view, similar to FIG. 10, of the operating device illustrated in FIGS. 8 and 11, but in which the grip cover has been removed.

[0069] FIG. 14 is a partial perspective view of a pommel portion of the operating device of illustrated in FIG. 13.

[0070] FIG. 15 is a partial exploded elevational view of the operating device of illustrated in FIGS. 13 and 14, but where an electrical unit has been detached from the base member.

[0071] FIG. 16 is a schematic diagram of the control system of the bicycle illustrated in FIG. 1.

[0072] FIG. 17 is a cross-sectional view of the electrical unit illustrated in FIGS. 15 and 16 as seen along section line 17-17 of FIG. 14.

[0073] FIG. 18 is a partial exploded perspective view of the electrical unit illustrated in FIGS. 15 to 17.

[0074] FIG. 19 is a perspective view of an alternate electrical unit in accordance with a second embodiment.

[0075] FIG. 20 is a perspective view of the electrical unit illustrated in FIG. 20 in which the battery is shown being inserted into the electrical unit.

[0076] FIG. 21 is a partial cross-sectional view of the electrical unit of illustrated in FIG. 20 as seen along section line 21-21 of FIG. 19.

[0077] FIG. 22 is a partial exploded perspective view of the electrical unit illustrated in FIGS. 19 to 21.

[0078] FIG. 23 is a partial exploded perspective view of an alternate electrical unit in accordance with a third embodiment.

[0079] FIG. 24 is a partial cross-sectional view of the electrical unit illustrated in FIG. 23.

[0080] FIG. 25 is a partial exploded perspective view of an alternate electrical unit in accordance with a fourth embodiment.

[0081] FIG. 26 is a partial cross-sectional view of the electrical unit illustrated in FIG. 25.

DETAILED DESCRIPTION

[0082] Selected embodiments will now be explained with reference to the drawings. It will be apparent to those skilled in the bicycle field from this disclosure that the following descriptions of the embodiments are provided for illustration only and not for the purpose of limiting the invention as defined by the appended claims and their equivalents.

[0083] Referring initially to FIGS. 1 and 2, a bicycle B is illustrated that is equipped with a control system 10 in accordance with illustrated embodiments. Here, in the first embodiment, the control system 10 basically includes at least two bicycle components. The term “bicycle component BC” as used herein can be used to generically refer to all of the bicycle components of the bicycle B. Here, in one configuration of the control system 10, the at least one bicycle component BC includes a first operating device 12, a second operating device 14, a first operated device 16, a second operated device 18, a third operated device 20 and a fourth operated device 22. In the first

embodiment, as explained below, the first operating device **12** is used to operate the first operated device **16**, the second operated device **18** and the third operated device **20**. Accordingly, the first operating device **12** will also be referred to as the bicycle component BC, the first operated device **16** will also be referred to as the first bicycle component BC1, the second operated device **18** will also be referred to as the second bicycle component BC2 and the third operated device **20** will also be referred to as the second bicycle component BC3.

[0084] It will be apparent from this disclosure that some of the bicycle components of the control system **10** can be omitted as needed and/or desired. Moreover, it will be apparent from this disclosure that other types of bicycle components can be added to the control system **10** as needed and/or desired. In addition, the bicycle component BC of the control system **10** is not limited to an operating device. Rather, the bicycle component BC of the control system **10** one of a gear changing device, a suspension, an adjustable seatpost, a brake device, a display device and a drive assist device.

[0085] Here, in the first embodiment, the control system **10** further includes a communication device CD. While FIG. **1** illustrates the communication device CD as a smartphone that is not mounted to the bicycle B, it will be apparent from this disclosure that the communication device CD can be mounted to the bicycle B as a cycle computer. Also, while the communication device CD is preferably a smartphone as shown, the communication device CD can be some other mobile device such as a smart watch, a wireless headphone, a tablet, a laptop, a cycle computer, etc. The communication device CD can also be referred to as an external device. The term “external device” as used herein refers to a device that is external with respect to the bicycle component BC. For example, an external device includes a cycle computer, a smartphone, a tablet, a personal computer, etc. Preferably, the external device includes a touch panel. In the case of a smartphone, for example, the external device can have a first function related to the bicycle B and a second function other than a function related to the bicycle B. For example, the first function of the external device can include one or more of diagnosing, adjusting, notifying, controlling and updating of a bicycle component, while the second function of the external device can include, for example, telephoning, e-mailing, web browsing, gaming, etc.

[0086] As shown in FIG. **1**, basically, the bicycle B includes a frame F that is supported by a rear wheel RW and a front wheel FW. A front suspension fork FF is pivotally coupled at its upper end to the frame F, and rotatably supports the front wheel FW at its lower end. The bicycle B further includes a handlebar HB mounted to the upper end of the front fork FF for steering the front wheel FW. The rear wheel RW is rotatably mounted to a rear end of the frame F. The seatpost SP is mounted to a seat tube of the frame F in a conventional manner and supports a bicycle seat or saddle S in any suitable manner. The bicycle B further includes a cycle computer CC mounted to the handlebar HB.

[0087] The bicycle B further includes a drivetrain DT. Here, for example, the drivetrain DT is a chain-drive type that includes a crank C, a plurality of front sprockets FS, a plurality of rear sprockets CS and a chain CN. The crank C includes a crank axle CA1 and a pair of crank arms CA2. The crank axle CA1 is rotatably supported to the front frame body FB via a bottom bracket in a conventional manner. The crank arms CA2 are provided on opposite ends of the crank axle CA1. A pedal PD is rotatably coupled to the distal end of each of the crank arms CA2. While the drivetrain DT is illustrated as a chain-drive type of drivetrain, the drivetrain DT can be selected from any type of drivetrain, and can be a belt-drive type or a shaft-drive type. The front sprockets FS are provided on the crank C to rotate integrally with the crank axle CA1. The rear sprockets CS are provided on a hub of the rear wheel RW. The chain CN runs around the front sprockets FS and the rear sprockets CS. A human driving force is applied to the pedals PD by a rider such that the driving force is transmitted via the front sprockets FS, the chain CN and the rear sprockets CS to the rear wheel RW.

[0088] As seen in FIG. **2**, the first operating device **12** and the second operating device **14** are

coupled to the handlebar HB in an installed state. Here, the handlebar HB is a drop handlebar. However, the first operating device **12** and the second operating device **14** can be reconfigured to be used with other types of handlebars, as needed and/or desired. The bicycle B has a center plane CP vertically bisecting the frame F of the bicycle B in a longitudinal direction (a front to rear direction). The center plane CP passes through a center of the frame F in a width direction (a left to right direction) of the frame F. Thus, the center plane CP separates a left side of the bicycle B from a right side of the bicycle B. The following directional terms “front,” “rear,” “forward,” “rearward,” “left,” “right,” “lateral,” “longitudinal,” “upward,” and “downward,” as well as any other similar directional terms, refer to those directions which are determined on the basis of a rider sitting upright on a seat of the bicycle B while facing the handlebar HB of the bicycle B.

[0089] Here, the first operating device **12** and the second operating device **14** are road briflers, which are operating devices that include both a braking function and a shifting function in a single unit that is mounted to the bicycle B. Here, the first operating device **12** is configured to at least control the first operated device **16** and the third operated device **20**. On the other hand, the second operating device **14** is configured to at least control the second operated device **18** and the fourth operated device **22**. Also, here, the first operated device **16** and the second operated device **18** are transmission devices, which are configured to change a gear ratio of the drivetrain DT of the bicycle B. The third operated device **20** and the fourth operated device **22** are brake devices, which are configured to apply a braking force to decelerate and/or stop rotation of either the rear wheel RW or the front wheel FW of the bicycle B.

[0090] Referring back to FIG. 1, in the first embodiment, the first operating device **12** is configured to wirelessly communicate with the first operated device **16** (e.g., a rear derailleur) and/or the second operated device **18** (e.g., a front derailleur). Alternately, the first operating device **12** can be configured to communicate with the first operated device **16** and/or the second operated device **18** via one or more wires. Likewise, the second operating device **14** can also be configured to wirelessly communicate with the first operated device **16** (e.g., a rear derailleur) and/or the second operated device **18** (e.g., a front derailleur). Alternately, the second operating device **14** can be configured to communicate with the first operated device **16** and/or the second operated device **18** via one or more wires. Thus, the first operating device **12** and the second operating device **14** can use wired communication and/or wireless communication, as needed and/or desired.

[0091] Here, the first operated device **16** is an electric rear derailleur, and the second operated device **18** is an electric front derailleur. The first operated device **16** (e.g., a rear derailleur) is configured to shift the chain CN between the rear sprockets CS in response to either an automatic shift signal from the cycle computer CC, or a user inputted shift signal from either the first operating device **12** or the second operating device **14**. The second operated device **18** (e.g., a front derailleur) is configured to shift the chain CN between the front sprockets FS in response to either an automatic shift signal from the cycle computer CC, or a user inputted shift signal either the first operating device **12** or the second operating device **14**.

[0092] In the first embodiment, the first operating device **12** and the second operating device **14** are configured to selectively operate in a full synchronized shift mode, a semi-synchronized shift mode and a full manual shift mode. In the full synchronized shift mode, the second operated device **18** (e.g., a front derailleur) is automatically shifted in response to the first operating device **12** being operated to shift the first operated device **16** (e.g., a rear derailleur). In the semi-synchronized shift mode, the first operated device **16** (e.g., a rear derailleur) is automatically shifted in response to the second operating device **14** being operated to shift the second operated device **18** (e.g., a front derailleur). In the full manual shift mode, the first operating device **12** is selectively operated to shift the first operated device **16** (e.g., a rear derailleur), and the second operating device **14** is selectively operated to shift the second operated device **18** (e.g., a front derailleur). Also, in a fully automatic transmission mode, the first operated device **16** (e.g., a rear derailleur) and the second operated device **18** (e.g., a front derailleur) can be automatically shifted without using either the

first operating device **12** or the second operating device **14**. For example, the first operated device **16** (e.g., a rear derailleur) and/or the second operated device **18** (e.g., a front derailleur) can be automatically shifted based on a cadence of the crank C and the torque applied to the crank arms CA2.

[0093] Still referring to FIG. 1, in the first embodiment, the first operating device **12** is operatively coupled to the third operated device **20** (e.g., a rear hydraulic brake device) via a hydraulic hose H1 to perform a braking operation. Similarly, as seen in FIG. 2, the second operating device **14** is operatively coupled to the fourth operated device **22** (e.g., a front hydraulic brake device) via a hydraulic hose H2 to perform a braking operation. The hydraulic hoses H1 and H2 are conventional structures commonly used in the bicycle field. Alternatively, the third operated device **20** and the fourth operated device **22** can be electric brake devices that are controlled by either wireless communication or wired communication as needed and/or desired.

[0094] Broadly speaking, the first operating device **12** and the second operating device **14** are used for selectively operating at least one other bicycle components of the bicycle B. The second operating device **14** is a mirror image of the first operating device **12**. Thus, the second operating device **14** will not be discussed in further detail. Rather, it will be apparent from this disclosure that the following description of the first operating device **12** also describes the second operating device **14**.

[0095] Referring now to FIGS. 3 to 11, the first operating device **12** will now be discussed in further detail. Basically, the bicycle component BC (the first operating device **12**) comprises a base member **30**. The base member **30** is configured to be mounted to the bicycle B. In the first embodiment, the base member **30** is configured to be coupled to the handlebar HB. In particular, the base member **30** includes a bracket or main body **32** and a mounting part **34**. The mounting part **34** is fixed to the main body **32**. The mounting part **34** is configured to be coupled to a curved portion of the handlebar HB. Of course, the first operating device **12** can be configured to be mounted on other portions of the bicycle B as needed and/or desired. Thus, the configuration of the body **32** and the mounting part **34** may be changed depending on where the bicycle component BC is mounted to the bicycle B.

[0096] Here, preferably, the main body **32** is made of a resin material. For example, the main body **32** is made of a hard-plastic material (resin), which can be reinforced with fibers as needed and/or desired. The resin of the main body **32** is a rigid material that is suitable for a rider to grip and lean on during riding. The resin of the main body **32** is also lightweight such that the overall weight of the first operating device **12** can be minimized. However, the main body **32** can be made of other suitable materials as needed and/or desired.

[0097] The main body **32** of the base member **30** includes a proximal end **32a** and a distal end **32b**. The proximal end **32a** is configured to be mounted to the bicycle B. The distal end **32b** is opposite to the proximal end **32a**. The proximal end **32a** is provided with the mounting part **34** for mounting to the handlebar HB. In particular, the mounting part **34** is attached to the proximal end **32a** of the main body **32** such that the mounting part **34** protrudes from the proximal end **32a** of the main body **32**. Thus, the proximal end **32a** is configured to be coupled to the handlebar HB by the mounting part **34**. The proximal end **32a** of the main body **32** can also be referred to a mounting end of the main body **32**. On the other hand, the distal end **32b** can also be referred to as a free end of the main body **32** in that the main body **32** is cantilever relative to the proximal end **32a** in a state where the bicycle component BC is mounted to the handlebar HB.

[0098] The mounting part **34** is configured to be mounted to the bicycle handlebar HB. In the first embodiment, the mounting part **34** is configured to be mounted to the right side of the handlebar HB. Alternatively, the mounting part **34** is configured to be mounted to the left side of the handlebar HB as needed and/or desired. The mounting part **34** is attached to the main body **32** such that the mounting part **34** is disposed in a handlebar receiving recess **32al** of the proximal end **32a**. The main body **32** is a stationary member when mounted to the handlebar HB by the mounting part

34. Since the main body **32** is fixed to the handlebar HB by the mounting part **34**, the base member **30** constitutes a fixed member with respect to the handlebar HB. The mounting part **34** is preferably a conventional band clamp or similar structure that is used in a road shifter for gripping the handlebar HB. Here, as seen in FIGS. **4** and **6**, the mounting part **34** includes a clamping band **36** and a fastener **38** (e.g., a nut and bolt) for gripping the handlebar HB. Since the mounting part **34** can be any suitable mounting structure, the mounting part **34** will not be discussed or illustrated in further detail herein.

[0099] In the first embodiment, the main body **32** of the base member **30** further includes a grip portion **32c** and a pommel portion **32d**. The grip portion **32c** extends from the proximal end **32a** to the distal end **32b**. The pommel portion **32d** is provided adjacent to the distal end **32b**. In other words, the pommel portion **32d** is part of the distal end portion of the base member **30**. The pommel portion **32d** is an upwardly protruding portion that protrudes upwardly relative to the grip portion **32c** in a state where the base member **30** is mounted to the curved portion of the handlebar HB. The grip portion **32c** is sized for the rider to grip and lean on during riding. Thus, the rider can grip the grip portion **32c** between the proximal end **32a** and the pommel portion **32d**.

[0100] As mentioned above, riders sometimes grip the main body **32** and lean on the main body **32** during riding. In view of this situation, it is desirable to provide a comfortable feeling for the rider's hand while the rider is gripping the main body **32**. Thus, the main body **32** is preferably covered with a grip cover **40** (also called a base cover). The grip cover **40** partially covers the main body **32** as seen in FIGS. **2** and **3**. Here, the grip cover **40** covers the grip portion **32c**, and also partially covers the pommel portion **32d**. Thus, the grip cover **40** provides a cushion to the grip portion **32c** of the base member **30**, and also provides an attractive appearance for the base member **30**. In this situation, the grip cover **40** is made of an elastomeric material. For example, the grip cover **40** is made of a flexible rubber material. The grip cover **40** has a tubular configuration such that the grip cover **40** is stretched over the base member **30**. In particular, the grip cover **40** is stretched over the grip portion **32c** of the main body **32** and the pommel portion **32d** of the main body **32**. In other words, the grip cover **40** is an elastic piece such as rubber that partially covers the outside surface of the base member **30** in the areas of the grip portion **32c** and the pommel portion **32d**.

[0101] Referring to FIGS. **3** and **4**, in the first embodiment, as mentioned above, the first operating device **12** is configured to be used for operating the first operated device **16** (e.g., a rear derailleur), the second operated device **18** (e.g., a front derailleur) and the third operated device **20** (e.g., a rear hydraulic brake device). In FIGS. **3** and **4**, the first operated device **16** (e.g., a rear derailleur) will be referred to as the first bicycle component BC1, the second operated device **18** (e.g., a front derailleur) will be referred to as the second bicycle component BC2, and the third operated device **20** (e.g., a rear brake device). Thus, in the first embodiment, as mentioned above, the first operating device **12** is a bicycle operating device that is configured to perform a braking operation and a shifting operation.

[0102] Referring now to FIGS. **3** to **11**, the bicycle component BC (e.g., the first operating device **12**) further comprises an operating member **42**. The operating member **42** is configured to operate an operated device mounted to the bicycle B. Here, the operating member **42** is configured to operate the first operated device **16** (e.g., a rear derailleur) and the second operated device **18** (e.g., a front derailleur). Basically, the operating member **42** is movably coupled relative to the base member **30**. In particular, the operating member **42** is configured to pivot relative to the base member **30** to operate at least one of the first operated device **16** (e.g., a rear derailleur) and the second operated device **18** (e.g., a front derailleur). More specifically, the operating member **42** operates at least one of the first operated device **16** (e.g., a rear derailleur) and the second operated device **18** (e.g., a front derailleur) in response to the operating member **42** being pivoted from a rest position (see FIG. **7**) to an operated position (see FIG. **8**). The operating member **42** is biased to the rest position. In other words, the operating member **42** automatically returns to the rest position upon the operating member **42** being released by the rider after the operating member **42** is

operated to perform the shifting operation.

[0103] The term “rest position” as used herein refers to a state in which a movable part (e.g., the operating member **42**) remains stationary without the need of a user or other external force intervening (e.g., holding the operating member **42**) to establish a state corresponding to the rest position. Thus, the term “rest position” can also be referred to as a non-operated position. The terms “operated position” and “actuated position” as used herein refer to a position at which the movable part (e.g., the operating member **42**) has been moved by a user from a rest position.

[0104] Here, in the first embodiment, the bicycle component BC (e.g., the first operating device **12**) further comprises another operating member **44** that is configured to operate an operated device mounted to the bicycle B. Here, the operating member **44** is configured to operate the first operated device **16** (e.g., a rear derailleur) and the second operated device **18** (e.g., a front derailleur). The operating member **42** can be referred to as the first operating member **42** or a first shift operating member **42** in the first embodiment. Also, the operating member **44** can be referred to as the second operating member **44** or a second shift operating member **44** in the first embodiment. Depending on the construction of the bicycle component BC (e.g., the first operating device **12**), the second operating member **44** can be omitted. In particular, the second operating member **44** is configured to pivot relative to the base member **30** to operate at least one of the first operated device **16** (e.g., a rear derailleur) and the second operated device **18** (e.g., a front derailleur). More specifically, the second operating member **44** operates at least one of the first operated device **16** (e.g., a rear derailleur) and the second operated device **18** (e.g., a front derailleur) in response to the second operating member **44** being pivoted from a rest position (see FIG. 7) to an operated position (see FIG. 9). The second operating member **44** is biased to the rest position. In other words, the second operating member **44** automatically returns to the rest position upon the second operating member **44** being released by the rider after the second operating member **44** is operated to perform the shifting operation.

[0105] In fully manual mode, the first operating member **42** can be used for upshifting the first operated device **16** (e.g., a rear derailleur) for shifting the chain CN to a smaller one of the sprockets CS. Also, in fully manual mode, the second operating member **44** can be used for downshifting the first operated device **16** (e.g., a rear derailleur) for shifting the chain CN to a larger one of the sprockets CS. In full synchronized shift mode, the first operated device **16** (e.g., a rear derailleur) and the second operated device **18** (e.g., a front derailleur) are in response to the first operating device **12** one of the first operating member **42** and the second operating member **44** being operated.

[0106] Here, in the first embodiment, the bicycle component BC (e.g., the first operating device **12**) further comprises an additional operating member **46**. The additional operating member **46** is movably coupled to the base member **30**. The additional operating member **46** is different from the operating member **42**. The additional operating member **46** is configured to operate an additional operated device mounted to the bicycle B. In the illustrated embodiment, the additional operating member **46** is configured to operate the third operated device **20** (e.g., a rear hydraulic brake device).

[0107] Here, as seen in FIGS. 10 and 11, the additional operating member **46** is a brake lever that is pivotally mounted to the main body **32** of the base member **30**. In particular, the additional operating member **46** is pivotally supported by an axle **48** to pivot with respect to the base member **30**. Here, the axle **48** is press-fitted into a bore provided in the main body **32** of the base member **30**. The axle **48** defines a first pivot axis P1. The axle **48** is disposed closer to the distal end **32b** than to the proximal end **32a**. The additional operating member **46** is configured to operate the third operated device **20** (e.g., a rear hydraulic brake device) in response to the additional operating member **46** being pivoted from a rest position (see FIG. 10) to an operated position (see FIG. 11). Thus, the additional operating member **46** can be referred to as the third operating member **46** or a brake operating member **46** in the first embodiment. The additional operating member **46** is biased

to the rest position. In other words, the additional operating member **46** automatically returns to the rest position upon the additional operating member **46** being released by the rider after the additional operating member **46** is operated to perform the braking operation.

[0108] Referring to FIGS. **5** to **9**, the operating member **42** is movably mounted to the additional operating member **46**. In particular, the first operating member **42** is pivotally mounted to the additional operating member **46** by an axle **50**. The axle **50** defines a second pivot axis **P2**. The second pivot axis **P2** is different from the first pivot axis **P1**. Here, the first pivot axis **P1** extends in a direction non-parallel to the second pivot axis **P2** as viewed from above in a direction perpendicular to the first pivot axis **P1** and the second pivot axis **P2**. More specifically, the first pivot axis **P1** extends perpendicularly to the second pivot axis **P2** as viewed from above in a direction perpendicular to the first pivot axis **P1** and the second pivot axis **P2**. Here, the axle **50** is a threaded fastener that is attached a rear side of the additional operating member **46**. In the first embodiment, the second operating member **44** is also pivotally mounted to the rear side of the additional operating member **46** by the axle **50**.

[0109] Still referring to FIGS. **5** to **9**, the basic operations of the first operating member **26** and the second operating member **44** will now be discussed. The bicycle component BC (e.g., the first operating device **12**) further comprises an electrical switch **52**. The electrical switch **52** can be a conventional electrical switch. Here, the electrical switch **52** provided with a first switch actuator that is operated by the first operating member **26** and a second switch actuator that is operated by the second operating member **44**. Alternatively, depending on the operating member, the electrical switch **52** can be provided with a single switch actuator as needed and/or desired. The electrical switch **52** is provided to at least one of the operating member **42** and the additional operating member **46**. In the first embodiment, the electrical switch **52** is attached a rear side of the additional operating member **46** by a pair of screws **54**. Thus, the electrical switch **52** is stationary with respect to the additional operating member **46**.

[0110] The operating member **42** is configured to activate the electrical switch **52** in response to an input to the operating member **42**. Likewise, the second operating member **44** is configured to activate the electrical switch **52** in response to an input to the second operating member **44**. Thus, the electrical switch **52** is selectively operated by user inputs to the first operating member **26** and/or the second operating member **44**. FIGS. **5** to **7** show the first operating member **26** and the second operating member **44** in their rest position.

[0111] In the case where a user input **IN1** is inputted to the first operating member **42** as seen in FIG. **8**, the first operating member **42** is pivoted about the second pivot axis **P2** from the rest position (see FIGS. **5** to **7**) to the operated position (see FIG. **8**). This movement of the first operating member **42** depresses a first actuator of the electrical switch **52** to output a first shift signal to perform a first shift operation. For example, in fully manual mode, the first shift signal is an upshift signal that is sent to the first operated device **16** (e.g., a rear derailleur) for shifting the chain **CN** to a smaller one of the sprockets **CS**. Thus, in fully manual mode, an upshift operation is performed in response to operation of the first operating member **42**. Also, when the first operating member **42** is pivoted about the second pivot axis **P2** from the rest position (see FIGS. **5** to **7**) to the operated position (see FIG. **8**), the second operating member **44** does not move, but remains in the rest position.

[0112] In the case where a user input **IN2** is inputted to the second operating member **44** as seen in FIG. **9**, the second operating member **44** is pivoted about the second pivot axis **P2** from the rest position (see FIGS. **5** to **7**) to the operated position (see FIG. **9**). This movement of the second operating member **44** depresses a second actuator of the electrical switch **52** to output a second shift signal to perform a second shift operation. For example, in fully manual mode, the second shift signal is a downshift signal that is sent to the first operated device **16** (e.g., a rear derailleur) for shifting the chain **CN** to a larger one of the sprockets **CS**. Thus, in fully manual mode, a downshift operation is performed in response to operation of the second operating member **44**.

Also, when the second operating member **44** is pivoted about the second pivot axis P2 from the rest position (see FIGS. 5 to 7) to the operated position (see FIG. 9), the first operating member **42** may move the second operating member **44**.

[0113] Referring to FIGS. 10 to 12, the basic operation of the additional operating member **46** (e.g., the brake lever) will now be discussed. As seen in FIG. 12, the bicycle component BC (e.g., the first operating device **12**) further comprises a hydraulic unit **56**. The hydraulic unit **56** is configured to operate the additional operated device in response to an input to the additional operating member **46**. Here, in the first embodiment, the additional operated device is the third operated device **20** (e.g., a rear hydraulic brake device). Thus, the additional operating member **46** (e.g., the brake lever) actuates the hydraulic unit **56** to operate the third operated device **20** (e.g., a rear hydraulic brake device).

[0114] Basically, the additional operating member **46** is configured to pivot between the rest position (FIG. 10) and an operated position (FIG. 11) in about the first pivot axis P1 to operate the third operated device **20** (e.g., a rear hydraulic brake device). In the case where a user input IN3 is inputted to the additional operating member **46** as seen in FIG. 11, the additional operating member **46** is pivoted about the first pivot axis P1 from the rest position (see FIG. 10) to the operated position (see FIG. 11). This movement of the additional operating member **46** actuates the hydraulic unit **56** to operate the third operated device **20** (e.g., a rear hydraulic brake device).

[0115] Here, in the first embodiment, this movement of the additional operating member **46** is configured to perform a braking operation. When the additional operating member **46** is pivoted to the operated position (FIG. 11), the first operating member **42** and the second operating member **44** move with the additional operating member **46**. However, the braking operation of the additional operating member **46** does not cause the electrical switch **52** to be actuated.

[0116] As seen in FIG. 12, the hydraulic unit **56** is integrated into the main body **32** of the base member **30**. In particular, the hydraulic unit **56** includes a cylinder bore **58** formed in the main body **32** of the base member **30**. The hydraulic unit **56** further comprises a piston **60**. The piston **60** is movably disposed in the cylinder bore **58**. The piston **60** moves linearly inside the cylinder bore **58** in response to a braking operation of the additional operating member **46**. Thus, the additional operating member **46** is operatively coupled to the piston **60** to move the piston **60** within the cylinder bore **58**. In particular, the hydraulic unit **56** further comprises an actuation part **62**. The actuation part **62** is operatively coupled between the piston **60** and the additional operating member **46**. The actuation part **62** has one end fixed to the additional operating member **46**, and in sliding contact with a pair of rollers **64** that are coupled to the piston **60**. In this way, the additional operating member **46** is connected to the piston **60** by the actuation part **62**.

[0117] The hydraulic unit **56** further comprises a piston biasing element **66** that is mounted to the main body **32** of the base member **30**. Specifically, the piston biasing element **66** biases the piston **60** to a non-actuated (rest or non-operated) position. Here, the piston biasing element **66** is a torsion spring that biases the piston **60** to the non-actuated or rest position. The piston biasing element **66** also biases the additional operating member **46** to its rest position (i.e., a state where no external force is applied to the additional operating member **46**). In the first embodiment, the hydraulic unit **56** further comprises a hydraulic reservoir **68** that is connected to the cylinder bore **58**. The hydraulic reservoir **68** is filled with a hydraulic fluid such as a mineral oil. The hydraulic reservoir **68** is disposed above the cylinder bore **58** where the first operating device **12** is in the mounted (installed) state. Since hydraulic units are well known, the hydraulic unit **56** will not be discussed in further detail for the sake of brevity.

[0118] Alternatively, the hydraulic unit **56** can be omitted and a conventional control cable can be attached to the additional operating member **46** for operating the third operated device **20** where the third operated device **20** is a mechanically operated brake device. Also, alternatively, the hydraulic unit **56** can be replaced with an electrical switch for operating the third operated device **20** where the third operated device **20** is an electrically operated brake device.

[0119] Referring now to FIGS. 13 to 18, in the first embodiment, the bicycle component BC (e.g., the first operating device 12) further comprises an electrical unit 70. Here, the electrical unit 70 forms a part of the pommel portion 32d of the main body 32. Basically, the electrical unit 70 is configured to supply electric power to the electrical switch 52 via an electrical cable 72, and to at least output control signals to the first operated device 16 (e.g., a rear derailleur). Here, the electrical unit 70 outputs control signals to the first operated device 16 (e.g., a rear derailleur), the second operated device 18 (e.g., a front derailleur), the cycle computer CC and the communication device CD.

[0120] In the first embodiment, the electrical unit 70 includes a holding structure 74 for holding various electrical parts that are provided to the base member 30. Preferably, the holding structure 74 is detachably attached to the pommel portion 32d of the main body 32 by a pair of screws. Thus, the bicycle component BC (e.g., the first operating device 12) further comprises the holding structure 74. Here, the holding structure 74 includes an accommodating part 76 and a power source cover 78. As seen in FIGS. 17 and 18, the power source cover 78 is pivotally coupled to the accommodating part 76 for accessing the electrical parts provided in the holding structure 74 of the electrical unit 70. Here, the holding structure 74 includes a cover fixing structure 80. The cover fixing structure 80 is configured to fix the power source cover 78 relative to the accommodating part 76. For example, as illustrated, the cover fixing structure 80 can be a latch that is pivotally mounted to the power source cover 78 and configured to engage a catch surface 76a of the accommodating part 76. The power source cover 78 can be provided with various sealing structures for preventing contaminants from entering the holding structure 74 between the accommodating part 76 and the power source cover 78 as needed and/or desired. For example, the power source cover 78 have a recess 78a for receiving an elastomeric O-ring 78b.

[0121] As seen in FIGS. 16 to 18, the bicycle component BC (e.g., the first operating device 12) further comprises a rechargeable power source 82. For example, the rechargeable power source 82 is a secondary battery. In the first embodiment, the rechargeable power source 82 includes at least one rechargeable coin battery. Alternatively, the rechargeable power source 82 can be a battery with a pluggable connector. Referring to FIG. 16, the rechargeable power source 82 is configured to be charged by electric power from an external power supply PS. Thus, the rechargeable power source 82 can be recharged as needed and/or desired.

[0122] The rechargeable power source 82 is configured to provide electric power to the electrical parts of the bicycle component BC (e.g., the first operating device 12). For example, the rechargeable power source 82 is configured to provide electric power to the electrical switch 52 via the electrical cable 72. To conserve power, as explained below, the rechargeable power source 82 is controlled to limit and/or stop the supply of electric power to the electrical switch 52 depending on an operating condition of the bicycle B or the bicycle component BC (e.g., the first operating device 12). In any case, the electrical switch 52 is configured to be activated by electric power from the rechargeable power source 82. In the first embodiment, the electrical switch 52 is configured to be activated by only electric power from the rechargeable power source 82. In one possible modification, the bicycle component BC (e.g., the first operating device 12) can include both a secondary battery and another battery (e.g., a primary battery) that can be used together to supply electric power to the electrical switch 52.

[0123] Here, the holding structure 74 is configured to removably hold the rechargeable power source 82. More specifically, the accommodating part 76 is configured to accommodate the rechargeable power source 82. The accommodating part 76 includes an opening 76b for removing the rechargeable power source 82 from the holding structure 74. In particular, in the first embodiment, the accommodating part 76 includes an accommodating space 76c for accommodating the rechargeable power source 82. Here, the accommodating space 76c is, for example, formed by a recess in an upper surface of the accommodating part 76. The opening 76b is formed by an upper edge of the accommodating space 76c. In the first embodiment, the

rechargeable power source **82** is supported on a holder **84**. Thus, the holder **84** supports the rechargeable power source **82** in the accommodating space **76c** of the accommodating part **76**. [0124] Referring to FIGS. **17** and **18**, the power source cover **78** is configured to at least partially cover the opening **76b** of the accommodating part **76**. Here, the power source cover **78** completely covers the opening **76b**, and is configured to restrict movement of the rechargeable power source **82** in the accommodating space **76c** of the accommodating part **76**. Alternatively, the power source cover **78** can be omitted and the grip cover **40** can serve as the power source cover as needed and/or desired. In the case where the grip cover **40** functions as the power source cover, then no other part is needed to cover the rechargeable power source **82**. In any case, the holding structure **74** is configured to detachably hold the rechargeable power source **82** to the bicycle component BC (e.g., the first operating device **12**) in the first embodiment.

[0125] Also, the bicycle component BC (e.g., the first operating device **12**) further comprises a substrate **86**. The substrate **86** has a first side **86a**, a second side **86b** and a peripheral edge **86c**. The second side **86b** faces in an opposite direction to the first side **86a**. The peripheral edge **86c** is disposed between the first side **86a** and the second side **86b**. The substrate **86** is provided in the holding structure **74**. Here, the holding structure **74** includes a pair of supports **74a** for supporting the rechargeable power source **82** and the substrate **86** in the accommodating space **76c** of the accommodating part **76**. The substrate **86** is a circuit board that supports various components. In first embodiment, the substrate **86** is a printed circuit board that is configured to electrically connect components to one another in a circuit.

[0126] The rechargeable power source **82** is electrically connected to the substrate **86** to provide electric power to the components that are provided to the substrate **86** or components that are electrically connected to the substrate **86**. Thus, the electrical switch **52** is configured to receive electric power from the rechargeable power source **82** via the electrical cable **72** and the substrate **86**. More specifically, the holding structure **74** includes an electrical terminal structure **88**. The electrical terminal structure **88** electrically couples the rechargeable power source **82** to the bicycle component BC (e.g., the first operating device **12**). In particular, the electrical terminal structure **88** is provided to the substrate **86** such that the electrical terminal structure **88** is electrically coupled to the substrate **86**. When the rechargeable power source **82** is installed in the holding structure **74**, the rechargeable power source **82** is electrically coupled to the substrate **86** via the electrical terminal structure **88**, and the electrical switch **52** is electrically coupled to the substrate **86** via the electrical cable **72**. In this way, the rechargeable power source **82** is electrically coupled to the electrical switch **52** via the electrical cable **72** and the substrate **86**. In the first embodiment, the electrical terminal structure **88** includes a first terminal **88A** and a second terminal **88B**. Here, the first terminal **88A** and the second terminal **88B** are electrical contacts that protrude from the substrate **86** for contacting the rechargeable power source **82**. Alternatively, the first terminal **88A** and the second terminal **88B** can be provided in a pluggable connector that is configured to receive a mating pluggable connector of a rechargeable power source. Also, alternatively, the first terminal **88A** and the second terminal **88B** can be electrical contacts that are contacted by terminals protruding from a button type rechargeable battery.

[0127] As seen in FIGS. **16** to **18**, the bicycle component BC (e.g., the first operating device **12**) further comprises a charging port **90**. The charging port **90** is electrically connected to the substrate **86**. Thus, the charging port **90** is electrically connected to the rechargeable power source **82** via the substrate **86**. In this way, the charging port **90** is configured to supply electrical power to the rechargeable power source **82** from the external power supply PS. In the first embodiment, the charging port **90** includes a USB Type connector. Alternatively, other types of electrical connectors can be used as needed and/or desired.

[0128] As seen in FIGS. **3**, **10** and **11**, the bicycle component BC (e.g., the first operating device **12**) further comprises a cover member **40a** configured to cover the charging port **90** in a case where the cover member **40a** is in a covering state. In the first embodiment, the cover member **40a** is a

part of the grip cover **40**. In particular, the cover member **40a** is a flap that is hinged to the rest of the grip cover **40** by a living hinge. The cover member **40a** covers a hole in the grip cover **40** and the opening of the charging port **90** in a case where the cover member **40a** is in a covering state. Alternatively, the cover member **40a** can be omitted such that the grip cover **40** forms the cover member for the charging port **90**. In this modification, the grip cover **40** can be removed from the pommel portion **32d** of the main body **32** to expose the opening of the charging port **90**.

[0129] Referring to FIGS. **16** and **18**, the bicycle component BC (e.g., the first operating device **12**) further comprises an electronic controller **92**. Basically, the electronic controller **92** includes at least one processor that is configured to execute predetermined control program (e.g., a pairing program, a shifting program, a power management program, etc.). The processor of the electronic controller **92** includes, for example, a central processing unit (CPU) or a micro processing unit (MPU). The electronic controller **92** is formed of one or more semiconductor chips that are mounted on the substrate **86**. Thus, the terms “controller” and “electronic controller” as used herein refer to hardware that executes a software program, and does not include a human being. The electronic controller **92** is configured to receive electric power from the rechargeable power source **82** via the substrate **86**. Preferably, the electronic controller **92** is configured to control the supply of electric power from the rechargeable power source **82** to the other electrical parts of the bicycle component BC (e.g., the first operating device **12**). The electronic controller **92** is preferably connected to a data storage device provided to the electrical unit **70**. For example, a data storage device can be provided on the substrate **86**. The data storage device can be any non-transitory computer readable medium such as a ROM (Read Only Memory) device, a RAM (Random Access Memory) device, a hard disk, a flash drive, etc. Here, for example, the data storage device includes, for example, at least one of a read-only memory (ROM), an erasable programmable read only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), and a flash memory. The data storage device stores various control processes or control programs, as well as information or data used by the electronic controller **92**.

[0130] The bicycle component BC (e.g., the first operating device **12**) further comprises a wireless communicator **94**. The wireless communicator **94** can be either a one-way wireless communicator (i.e., a wireless transmitter) or a two-way wireless communicator (i.e., a wireless transceiver) as needed and/or desired. For example, in the first embodiment, the wireless communicator **94** includes a signal transmitting circuit (TX circuit) and a signal receiving circuit (RX circuit) so that the wireless communicator **94** is configured to both receive and transmit wireless communication signals, or just a signal transmitting circuit (TX circuit).

[0131] The substrate **86** supports the wireless communicator **94**. In particular, in the first embodiment, the wireless communicator **94** is electrically connected to the electronic controller **92** via the substrate **86**. The electronic controller **92** is configured to control the wireless communicator **94**. Alternatively, the wireless communicator **94** can be provided with its own electronic controller as needed and/or desired. Also, the wireless communicator **94** is electrically coupled to the rechargeable power source **82**. Here, the wireless communicator **94** is configured to receive electric power from the rechargeable power source **82** via the substrate **86**. Preferably, the electronic controller **92** is configured to control the supply of electric power from the rechargeable power source **82** to the wireless communicator **94**.

[0132] The bicycle component BC (e.g., the first operating device **12**) further comprises an antenna **96**. The antenna **96** is configured to receive and transmit wireless signals from the wireless communicator **94**. The antenna **96** is provided to the substrate **86** and electrically coupled to the wireless communicator **94**. A signal amplifier can be provided to the substrate **86** for selectively amplifying the signals of the antenna **96** as needed and/or desired. The substrate **86** is disposed between the rechargeable power source **82** and the antenna **96**. In particular, in the first embodiment, the substrate **86** supports the wireless communicator **94** on the first side **86a**. The antenna **96** is provided to the second side **86b** of the substrate **86**, and is electrically coupled to the

wireless communicator **94**. The rechargeable power source **82** is closer to the power source cover **78** than the antenna **96**.

[0133] The electronic controller **92** is configured to control supply of electrical power from the rechargeable power source **82** to the wireless communicator **94** based on an output from a detector **98**. The detector **98** is configured to detect a state of at least one of the bicycle B and the bicycle component BC (e.g., the first operating device **12**). For example, the detector **98** is a wake up detector that is configured to detect a vibration, and/or a movement of the bicycle B or the bicycle component BC that is indicative of the bicycle B and/or the bicycle component BC being used. The detector **98** can be referred to as a first detector, or as a wake up detector. As shown in FIG. **16**, in the first embodiment, the wake up detector **98** is provided on the substrate **86**. However, the location of the wake up detector **98** is not limited to the illustrated location. Here, the wake up detector **98** can be an acceleration sensor that detects a vibration, and/or a movement of the bicycle B or the bicycle component BC that is indicative of the bicycle B and/or the bicycle component BC being used.

[0134] As shown in FIG. **16**, in the first embodiment, a second detector **99** is provided in the electrical unit **70**. The second detector **99** is configured to measure at least one of voltage, current, temperature, and other properties of the rechargeable power source **82**. The second detector **99** can be referred to as a SOC detector. The electronic controller **92** is configured to determine a current state of charge of the rechargeable power source **82** from the detected values of the second detector **99**. Here, the substrate **86** is further provided with the second detector **99**. However, the location of the second detector **99** is not limited to the illustrated location.

[0135] The electronic controller **92** is configured to obtain information related to the rechargeable power source **82**. For example, the electronic controller **92** is configured to obtain a current power level or state of charge (SOC) of the rechargeable power source **82** from the second detector **99**. Also, for example, the electronic controller **92** is configured to obtain a battery life (e.g., a number of charge and discharge cycles that can be completed before losing performance) of the rechargeable power source **82** from the second detector **99**. The wireless communicator **94** is configured to transmit the information related to the rechargeable power source **82** to an external device. For example, the wireless communicator **94** is configured to transmit a current power level or state of charge (SOC) of the rechargeable power source **82** to the external device (e.g., the communication device CD or the cycle computer CC). Also, for example, the wireless communicator **94** is configured to transmit a battery life (e.g., a number of charge and discharge cycles that can be completed before losing performance) to the external device (e.g., the communication device CD or the cycle computer CC).

[0136] As shown in FIG. **16**, in the first embodiment, the bicycle component BC (e.g., the first operating device **12**) further comprises a notification device **100** electrically coupled to the electronic controller **92**. Here, for example, the notification device **100** includes an LED circuit and a light transmission member that is configured to transmit light emitted from the LEDs of the LED circuit. Here, the LED circuit of the notification device **100** is provided on the second side **86b** of the substrate **86**. However, the notification device **100** can be provided at a location remote from the substrate **86**. The light transmission member of the notification device **100** extends from a position adjacent to the LEDs of the LED circuit to a window portion formed in the exterior of the electrical unit **70**. The electronic controller **92** is configured to control the notification device **100** based on the information. For example, the LED circuit of the notification device **100** is controlled by the electronic controller **92** such that one or more LEDs are illuminated to produce a desired notification (e.g., blue light, a red light, a green light, a steady state light, a flashing light, etc.) to indicate the information that was determined based on the second detector **99**.

[0137] Referring back to FIG. **16**, the electrical configurations of the first operated device **16** (e.g., a rear derailleur) and the second operated device **18** (e.g., a front derailleur) will now be discussed. For example, the first operated device **16** (e.g., a rear derailleur) and the second operated device **18**

(e.g., a front derailleur) each includes an electronic controller **110** and a wireless communicator **112**. Also, the first operated device **16** (e.g., a rear derailleur) and the second operated device **18** (e.g., a front derailleur) each preferably includes a position detector **114**, an actuator **116**, an actuator driver **118**, a power supply **120** and an antenna **122**. Here, the electronic controller **110**, the wireless communicator **112**, the position detector **114** and the antenna **122** are provided to a circuit board **124**. However, it will be apparent from this disclosure that the electronic controller **110**, the wireless communicator **112**, the position detector **114** and the antenna **122** can be provided at different locations to different circuit boards.

[0138] Also, while the first operated device **16** (e.g., a rear derailleur) and the second operated device **18** (e.g., a front derailleur) are illustrated with their own individual power supply, the first operated device **16** (e.g., a rear derailleur) and the second operated device **18** (e.g., a front derailleur) can share a power supply that is provided to one of the bicycle components BC or remotely located on the frame F of the bicycle B.

[0139] The wireless communicator **112** receives wireless control signals transmitted from the wireless communicator **84** of the first operating device **12** or the second operating device **14**. The wireless control signals are processed by the electronic controller **110**, which controls the actuator **116** via the actuator driver **118**. Thus, the actuator driver **118** control the actuator **116** based on the control signal of the electronic controller **110**.

[0140] Examples of the actuator **116** include a direct-current (DC) motor and a stepper motor. In the case where the first operated device **16** and the second operated device **18** are external transmission devices (e.g., derailleurs), the first operated device **16** and the second operated device **18** are provided a chain guide, and the actuator **116** is operatively coupled to the chain guide more moving the chain guide. The position detector **114** is configured to sense a current gear position of the rear derailleur **14**. Examples of the position detector **114** include a potentiometer and a rotary encoder. The position detector **114** is configured to sense an absolute rotational position of a rotational shaft of the actuator **116** as the current gear position of the gear rear derailleur **14**. The actuator **116** and the position detector **114** are electrically connected to the actuator driver **118**.

[0141] Referring now to FIGS. **19** to **22**, an electrical unit **270** is illustrated in accordance with a second embodiment. The electrical unit **270** is configured to be used with the base member **30** of the first embodiment. Thus, the electrical unit **270** is configured to be electrically connected to the electrical switch **52** via an electrical cable **272**. The electrical unit **270** is functionally identical to the electrical unit **70**. The electrical unit **270** is an alternate configuration of the parts from the configuration of the parts of the electrical unit **70**. In view of the functional similarity between the first embodiment and the second embodiment, the following the descriptions of the second embodiment will focus on the differences in the configuration of the electrical unit **270** from the configuration of the electrical unit **70**.

[0142] Basically, the electrical unit **270** includes a holding structure **274** for holding various electrical parts that are provided to the base member **30**. Preferably, the holding structure **274** is detachably attached to the pommel portion **32d** of the main body **32** by a pair of screws in the same manner as the first embodiment.

[0143] In the second embodiment, as seen in FIGS. **19** to **22**, the holding structure **274** includes an accommodating part **276** and a power source cover **278**. Here, the accommodating part **276** includes an upper portion and a lower portion that are snap-fitted together. The power source cover **278** is pivotally coupled to the upper portion of the accommodating part **276**. Here, the holding structure **274** includes a cover fixing structure **280**. For example, as illustrated, the cover fixing structure **280** can be a latch that is provided to the power source cover **278** and configured to engage a catch surface **276a** of the lower portion of the accommodating part **276**. Preferably, the catch surface **276a** is formed by a notch **277** in the accommodating part **276**. The notch **277** is sized to receive a tool such as a screwdriver for disengaging the cover fixing structure **280** (e.g., the latch) from the catch **276a**.

[0144] The electrical unit **270** further comprises a rechargeable power source **282**. For example, the rechargeable power source **282** is a secondary battery. In the second embodiment, the rechargeable power source **282** includes at least one rechargeable coin battery. Alternatively, the rechargeable power source **282** can be a battery with a pluggable connector. The rechargeable power source **282** is configured to provide electric power to the electrical parts of the electrical unit **270** and the electrical switch **52** where the electrical unit **270** is installed to the base member **30**. Thus, the electrical switch **52** is configured to be activated by electric power from the rechargeable power source **282** where the electrical unit **270** is installed to the base member **30**. In the second embodiment, the electrical switch **52** is configured to be activated by only electric power from the rechargeable power source **282**.

[0145] In the second embodiment, the holding structure **274** is configured to removably hold the rechargeable power source **282**. More specifically, the accommodating part **276** is configured to accommodate the rechargeable power source **282**. The accommodating part **276** includes an opening **276b** for removing the rechargeable power source **282** from the holding structure **274**. The opening **276b** is in an upper portion of the accommodating part **276**. Also, in the second embodiment, the accommodating part **276** includes an accommodating space **276c** for accommodating the rechargeable power source **282** after the rechargeable power source **282** has been inserted through the opening **276b**. In the second embodiment, the rechargeable power source **282** is supported on a holder **284** provided in the accommodating space **276c** of the accommodating part **276**. Thus, the holder **284** supports the rechargeable power source **282** in the accommodating space **276c** of the accommodating part **276** so that the rechargeable power source **282** rests on the holder **284** after the rechargeable power source **282** has been inserted through the opening **276b**. Preferably, the holding structure **274** includes a battery removal aid such as a battery removal ribbon to aid in the removal of the rechargeable power source **282**. For example, one end of the battery removal ribbon is attached to the holder **284** and the battery removal ribbon wraps around the end of the battery opposite to the opening **276b** so that a user can pull battery removal ribbon to pop the rechargeable power source **282** out through the opening **276b**.

[0146] The electrical unit **270** further comprises a substrate **286**. The substrate **286** has a first side **286a**, a second side **286b** and a peripheral edge **286c**. The second side **286b** faces in an opposite direction to the first side **286a**. The peripheral edge **286c** is disposed between the first side **286a** and the second side **286b**. The substrate **286** is provided in the holding structure **274**. In particular, the substrate **286** is provided on the upper surface of the holder **284** in the accommodating space **276c** of the accommodating part **276**. The holder **284** supports the substrate **286** in the accommodating space **276c** of the accommodating part **276**. The substrate **286** is a circuit board that supports various components. In the second embodiment, the substrate **286** is a printed circuit board that is configured to electrically connect components to one another in a circuit.

[0147] The rechargeable power source **282** is electrically connected to the substrate **286** to provide electric power to the components that are provided to the substrate **286** or components that are electrically connected to the substrate **286**. More specifically, the holding structure **274** includes an electrical terminal structure **288** that is provided to the substrate **286**. The electrical terminal structure **288** contacts the rechargeable power source **282** to electrically couple the rechargeable power source **282** to the substrate **286**. Thus, the electrical switch **52** is configured to receive electric power from the rechargeable power source **282** via the electrical cable **272** and the substrate **286**.

[0148] In the second embodiment, the electrical terminal structure **288** includes a first or terminal **288A** and a second terminal **288B**. Here, the first terminal **288A** and the second terminal **288B** are electrical contacts that protrude from the substrate **286** for contacting the rechargeable power source **282**. Alternatively, the first terminal **288A** and the second terminal **288B** can be provided in a pluggable connector that is configured to receive a mating pluggable connector of a rechargeable power source. Also, alternatively, the first terminal **288A** and the second terminal **288B** can be

electrical contacts that are contacted by terminals protruding from a button type rechargeable battery.

[0149] The electrical unit **270** further comprises a charging port **290**. The charging port **290** is electrically connected to the substrate **286**. Thus, the charging port **290** is electrically connected to the rechargeable power source **282** via the substrate **286**. In this way, similar to the first embodiment, the charging port **290** is configured to supply electrical power to the rechargeable power source **282** from an external power supply. In the second embodiment, the charging port **290** includes a USB Type connector. Alternatively, other types of electrical connectors can be used as needed and/or desired.

[0150] The electrical unit **270** further comprises an electronic controller **292**. Basically, the electronic controller **292** includes at least one processor that is configured to execute predetermined control program (e.g., a pairing program, a shifting program, a power management program, etc.). The processor of the electronic controller **292** includes, for example, a central processing unit (CPU) or a micro processing unit (MPU). The electronic controller **292** is formed of one or more semiconductor chips that are mounted on the substrate **286**. The electronic controller **292** has the same configuration of the electronic controller **92**, which is discussed above, and thus, the electronic controller **292** will not be discussed in further detail.

[0151] The electrical unit **270** further comprises a wireless communicator **294**. The wireless communicator **294** can be either a one-way wireless communicator (i.e., a wireless transmitter) or a two-way wireless communicator (i.e., a wireless transceiver) as needed and/or desired. The wireless communicator **294** has the same configuration of the wireless communicator **94**, which is discussed above, and thus, the wireless communicator **294** will not be discussed in further detail. Here, the substrate **286** supports the wireless communicator **294**, and electrically connects the wireless communicator **294** to the electronic controller **292** via the substrate **286**. Alternatively, the wireless communicator **294** can be provided with its own electronic controller as needed and/or desired. The wireless communicator **294** is configured to receive electric power from the rechargeable power source **282** via the substrate **286**. Preferably, the electronic controller **292** is configured to control the supply of electric power from the rechargeable power source **282** to the wireless communicator **294**.

[0152] The electrical unit **270** further comprises an antenna **296**. The antenna **296** is configured to receive and transmit wireless signals from the wireless communicator **294**. The antenna **296** is provided to the substrate **286** and electrically coupled to the wireless communicator **294**. In the second embodiment, the antenna **296** is disposed along a peripheral edge **286c** of the substrate **286**. A signal amplifier can be provided to the substrate **286** for selectively amplifying the signals of the antenna **296** as needed and/or desired. The substrate **286** is disposed between the rechargeable power source **282** and the antenna **296**.

[0153] The electronic controller **292** is configured to control supply of electrical power from the rechargeable power source **282** to the wireless communicator **294** based on an output from a detector **298**. The detector **298** is configured to detect a state of at least one of the bicycle B and the bicycle component BC (e.g., the first operating device **12**). For example, the detector **298** is a wake up detector that is configured to detect a vibration, and/or a movement of the bicycle B or the bicycle component BC that is indicative of the bicycle B and/or the bicycle component BC being used. As shown in FIG. 22, in the second embodiment, the wake up detector **298** is provided on the substrate **286**. However, the location of the wake up detector **298** is not limited to the illustrated location. Here, the wake up detector **298** can be an acceleration sensor that detects a vibration, and/or a movement of the bicycle B or the bicycle component BC that is indicative of the bicycle B and/or the bicycle component BC being used.

[0154] The electrical unit **270** also further comprises a SOC detector and a notification device similar to the SOC detector and a notification device of the electrical unit **70**. In the second embodiment, the SOC detector and the notification device are identical to the second (SOC)

detector **99** and the notification device **100**, which are discussed above, and thus, the SOC detector and the notification device of the electrical unit **270** will not be further discussed.

[0155] Referring now to FIGS. **23** and **24**, an electrical unit **370** is illustrated in accordance with a third embodiment. The electrical unit **370** is configured to be used with the base member **30** of the first embodiment. Thus, the electrical unit **370** is configured to be electrically connected to the electrical switch **52** via an electrical cable **372**. The electrical unit **370** is functionally identical to the electrical unit **70**. The electrical unit **370** is an alternate configuration of the parts from the configuration of the parts of the electrical unit **70**. In view of the functional similarity between the third embodiment and the prior embodiments, the following the descriptions of the third embodiment will focus on the differences in the configuration of the electrical unit **370** from the configuration of the electrical unit **70**.

[0156] Basically, the electrical unit **370** includes a holding structure **374** for holding various electrical parts that are provided to the base member **30**. Preferably, the holding structure **374** is detachably attached to the pommel portion **32d** of the main body **32** by a pair of screws in the same manner as the first embodiment. The holding structure **374** is basically the same as the second embodiment. The holding structure **374** includes an accommodating part **376** and a power source cover **378**. The power source cover **378** is pivotally coupled to the upper portion of the accommodating part **376**. Here, the holding structure **374** includes a cover fixing structure **380**. For example, as illustrated, the cover fixing structure **380** can be a latch that is provided to the power source cover **378** and configured to engage a catch surface **376a** of the lower portion of the accommodating part **376**. Preferably, the catch surface **376a** is formed by a notch **377** in the accommodating part **376**. The notch **377** is sized to receive a tool such as a screwdriver for disengaging the cover fixing structure **380** (e.g., the latch) from the catch **376a**.

[0157] The holding structure **374** is configured to removably hold the rechargeable power source **382** similar to the second embodiment. The accommodating part **376** includes an opening **376b** for removing the rechargeable power source **382** from the holding structure **374**. The opening **376b** is in an upper portion of the accommodating part **376**. Also, the accommodating part **376** includes an accommodating space **376c** for accommodating the rechargeable power source **382** after the rechargeable power source **382** has been inserted through the opening **376b**. In the third embodiment, the rechargeable power source **382** is supported on a holder **384** so that the rechargeable power source **382** rests on the holder **384** after the rechargeable power source **382** has been inserted through the opening **376b**.

[0158] The electrical unit **370** further comprises a substrate **386** having a first side **386a**, a second side **386b** and a peripheral edge **386c**. The substrate **386** is provided in the holding structure **374**. In particular, the substrate **386** is provided on the upper surface of the holder **384** in the accommodating space **376c** of the accommodating part **376**. The holder **384** supports the substrate **386** in the accommodating space **376c** of the accommodating part **376**. The substrate **386** is a circuit board that supports various components. Here, the substrate **386** is a printed circuit board that is configured to electrically connect components to one another in a circuit.

[0159] Here, the rechargeable power source **382** is a rectangular battery that is pushed through the opening **376b** into the accommodating part **376**. Preferably, a latch is provided in the accommodating part **376** to prevent movement of the rechargeable power source **382** in the accommodating space **376c** of the accommodating part **376**. The rechargeable power source **382** is electrically connected to the substrate **386** to provide electric power to the components that are provided to the substrate **386** or components that are electrically connected to the substrate **386**. More specifically, the holding structure **374** includes an electrical terminal structure **388** that is provided to the substrate **386**. The electrical terminal structure **388** contacts electrical contacts **382a** and **382b** of the rechargeable power source **382** to electrically couple the rechargeable power source **382** to the substrate **386**. Thus, the electrical switch **52** is configured to receive electric power from the rechargeable power source **382** via the electrical cable **372** and the substrate **386**.

[0160] In the third embodiment, the electrical terminal structure **388** includes a first or terminal **388A** and a second terminal **388B**. Here, the first terminal **388A** and the second terminal **388B** are electrical contacts that extend along the peripheral edge **386c** of the substrate **386** for contacting the electrical contacts **382a** and **382b** of the rechargeable power source **382**.

[0161] The electrical unit **370** further comprises a charging port **390**. The charging port **390** is electrically connected to the substrate **386**. Thus, the charging port **390** is electrically connected to the rechargeable power source **382** via the substrate **386**. In this way, similar to the first embodiment, the charging port **390** is configured to supply electrical power to the rechargeable power source **382** from an external power supply.

[0162] The electrical unit **370** further comprises an electronic controller **392**. Basically, the electronic controller **392** includes at least one processor that is configured to execute predetermined control program (e.g., a pairing program, a shifting program, a power management program, etc.). The processor of the electronic controller **392** includes, for example, a central processing unit (CPU) or a micro processing unit (MPU). The electronic controller **392** is formed of one or more semiconductor chips that are mounted on the substrate **386**. The electronic controller **392** has the same configuration of the electronic controller **92**, which is discussed above, and thus, the electronic controller **392** will not be discussed in further detail.

[0163] The electrical unit **370** further comprises a wireless communicator **394**. The wireless communicator **394** can be either a one-way wireless communicator (i.e., a wireless transmitter) or a two-way wireless communicator (i.e., a wireless transceiver) as needed and/or desired. The wireless communicator **394** has the same configuration of the wireless communicator **94**, which is discussed above, and thus, the wireless communicator **394** will not be discussed in further detail. Here, the substrate **386** supports the wireless communicator **394**, and electrically connects the wireless communicator **394** to the electronic controller **392** via the substrate **386**. The wireless communicator **394** is configured to receive electric power from the rechargeable power source **382** via the substrate **386**. Preferably, the electronic controller **392** is configured to control the supply of electric power from the rechargeable power source **382** to the wireless communicator **394**.

[0164] The electrical unit **370** further comprises an antenna **396**. The antenna **396** is configured to receive and transmit wireless signals from the wireless communicator **394**. The antenna **396** is provided to the substrate **386** and electrically coupled to the wireless communicator **394**. Here, the antenna **396** is disposed along a peripheral edge **386c** of the substrate **386**. A signal amplifier can be provided to the substrate **386** for selectively amplifying the signals of the antenna **396** as needed and/or desired.

[0165] The electronic controller **392** is configured to control supply of electrical power from the rechargeable power source **382** to the wireless communicator **394** based on an output from a detector **398**. The detector **398** is configured to detect a state of at least one of the bicycle B and the bicycle component BC (e.g., the first operating device **12**). For example, the detector **398** is a wake up detector that is configured to detect a vibration, and/or a movement of the bicycle B or the bicycle component BC that is indicative of the bicycle B and/or the bicycle component BC being used. Here, in the third embodiment, the wake up detector **398** is provided on the substrate **386**. The electrical unit **370** also further comprises a SOC detector and a notification device similar to the SOC detector and a notification device of the electrical unit **70**.

[0166] Referring now to FIGS. 25 and 26, an electrical unit **470** is illustrated in accordance with a fourth embodiment. The electrical unit **470** is configured to be used with the base member **30** of the first embodiment. Thus, the electrical unit **470** is configured to be electrically connected to the electrical switch **52** via an electrical cable **472**. The electrical unit **70** is functionally identical to the electrical unit **70**. The electrical unit **470** is an alternate configuration of the parts from the configuration of the parts of the electrical unit **70**. In view of the functional similarity between the fourth embodiment and the prior embodiments, the following the descriptions of the fourth embodiment will focus on the differences in the configuration of the electrical unit **470** from the

configuration of the electrical unit **70**.

[0167] Basically, the electrical unit **470** includes a holding structure **474** for holding various electrical parts that are provided to the base member **30**. Preferably, the holding structure **474** is detachably attached to the pommel portion **32d** of the main body **32** by a pair of screws in the same manner as the first embodiment. The holding structure **474** is basically the same as the first embodiment. The holding structure **474** includes an accommodating part **476** and a power source cover **478**. The power source cover **478** is pivotally coupled to the upper portion of the accommodating part **476**. Here, the holding structure **474** includes a cover fixing structure **480**. For example, as illustrated, the cover fixing structure **480** can be a latch that is provided to the power source cover **478** and configured to engage a catch surface **476a** of the accommodating part **476**.

[0168] The holding structure **474** is configured to removably hold the rechargeable power source **482** similar to the first embodiment. The accommodating part **476** includes an opening **476b** for removing the rechargeable power source **482** from the holding structure **474**. The opening **476b** is in an upper portion of the accommodating part **476**. Also, the accommodating part **476** includes an accommodating space **476c** for accommodating the rechargeable power source **482** after the rechargeable power source **482** has been inserted through the opening **476b**. In the fourth embodiment, the rechargeable power source **482** is supported on a holder **484** provided in the accommodating space **476c** of the accommodating part **476**.

[0169] The electrical unit **470** further comprises a substrate **486** having a first side **486a**, a second side **486b** and a peripheral edge **486c**. The substrate **486** is provided in the holding structure **474**. In particular, the substrate **486** is provided on the upper surface of the holder **484** in the accommodating space **476c** of the accommodating part **476**. The holder **484** supports the substrate **486** in the accommodating space **476c** of the accommodating part **476**. The substrate **486** is provided in the holding structure **474**. Here, the holding structure **474** includes a pair of supports **474a** for supporting the rechargeable power source **482** and the substrate **486** in the accommodating space **476c** of the accommodating part **476**. The substrate **486** is a circuit board that supports various components. Here, the substrate **486** is a printed circuit board that is configured to electrically connect components to one another in a circuit.

[0170] Here, the rechargeable power source **482** is a rectangular battery that is provided with a pluggable electrical connector **482a**. The rechargeable power source **482** is electrically connected to the substrate **486** to provide electric power to the components that are provided to the substrate **486** or components that are electrically connected to the substrate **486**. More specifically, the holding structure **474** includes an electrical terminal structure **488** that is provided to the substrate **486**. The electrical terminal structure **488** contacts the rechargeable power source **482** to electrically couple the rechargeable power source **482** to the substrate **486**. Thus, the electrical switch **52** is configured to receive electric power from the rechargeable power source **482** via the electrical cable **472** and the substrate **486**.

[0171] In the fourth embodiment, the electrical terminal structure **488** is a pluggable electrical connector that is mounted on the first side **486a** of the substrate **486**. Here, the pluggable electrical connector **482a** of the rechargeable power source **482** is configured to mate with the electrical terminal structure **488** for electrically connecting the rechargeable power source **482** to the circuitry of the substrate **486**. In other words, the pluggable electrical connector **482a** of the rechargeable power source **482** is pushed into connection with the pluggable electrical connector of the electrical terminal structure **488**. For example, the pluggable electrical connector **482a** of the rechargeable power source **482** is a male electrical connector and the pluggable electrical connector of the electrical terminal structure **488** is a female electrical connector.

[0172] The electrical unit **470** further comprises a charging port **490**. The charging port **490** is electrically connected to the substrate **486**. Thus, the charging port **490** is electrically connected to the rechargeable power source **482** via the substrate **486**. In this way, similar to the first embodiment, the charging port **490** is configured to supply electrical power to the rechargeable

power source **482** from an external power supply.

[0173] The electrical unit **470** further comprises an electronic controller **492**. Basically, the electronic controller **492** includes at least one processor that is configured to execute predetermined control program (e.g., a pairing program, a shifting program, a power management program, etc.). The processor of the electronic controller **492** includes, for example, a central processing unit (CPU) or a micro processing unit (MPU). The electronic controller **492** is formed of one or more semiconductor chips that are mounted on the substrate **486**. The electronic controller **492** has the same configuration of the electronic controller **92**, which is discussed above, and thus, the electronic controller **492** will not be discussed in further detail.

[0174] The electrical unit **470** further comprises a wireless communicator **494**. The wireless communicator **494** can be either a one-way wireless communicator (i.e., a wireless transmitter) or a two-way wireless communicator (i.e., a wireless transceiver) as needed and/or desired. The wireless communicator **494** has the same configuration of the wireless communicator **94**, which is discussed above, and thus, the wireless communicator **494** will not be discussed in further detail. Here, the substrate **486** supports the wireless communicator **494**, and electrically connects the wireless communicator **494** to the electronic controller **492** via the substrate **486**. The wireless communicator **494** is configured to receive electric power from the rechargeable power source **482** via the substrate **486**. Preferably, the electronic controller **492** is configured to control the supply of electric power from the rechargeable power source **482** to the wireless communicator **494**.

[0175] The electrical unit **470** further comprises an antenna **496**. The antenna **496** is configured to receive and transmit wireless signals from the wireless communicator **494**. The antenna **496** is provided to the substrate **486** and electrically coupled to the wireless communicator **494**. Here, the antenna **496** is provided to the second side **486b** of the substrate **486**, and is electrically coupled to the wireless communicator **494**. A signal amplifier can be provided to the substrate **486** for selectively amplifying the signals of the antenna **496** as needed and/or desired.

[0176] The electronic controller **492** is configured to control supply of electrical power from the rechargeable power source **482** to the wireless communicator **494** based on an output from a detector **498**. The detector **498** is configured to detect a state of at least one of the bicycle B and the bicycle component BC (e.g., the first operating device **12**). For example, the detector **498** is a wake up detector that is configured to detect a vibration, and/or a movement of the bicycle B or the bicycle component BC that is indicative of the bicycle B and/or the bicycle component BC being used. Here, in the third embodiment, the wake up detector **498** is provided on the substrate **486**. The electrical unit **470** also further comprises a SOC detector and a notification device similar to the SOC detector and a notification device of the electrical unit **70**.

[0177] In the prior embodiments, the electrical units **70**, **270**, **370** and **470** can be modified such that the rechargeable power sources **82**, **282**, **382** and **482** can be charged wirelessly. In the modification where the rechargeable power sources **82**, **282**, **382** and **482** can be charged wirelessly, the rechargeable power sources **82**, **282**, **382** and **482** are provided so as to be not detachably from the electrical units **70**, **270**, **370** and **470**. For example, in the first embodiment, the power source cover **78** can be fixed to the accommodating part **76** by one or more fixing bolts. Also, for example, in the second embodiment, the power source cover **278** and the opening **276b** can be omitted.

[0178] In understanding the scope of the present invention, the term “comprising” and its derivatives, as used herein, are intended to be open ended terms that specify the presence of the stated features, elements, components, groups, integers, and/or steps, but do not exclude the presence of other unstated features, elements, components, groups, integers and/or steps. The foregoing also applies to words having similar meanings such as the terms, “including”, “having” and their derivatives. Also, the terms “part,” “section,” “portion,” “member” or “element” when used in the singular can have the dual meaning of a single part or a plurality of parts unless otherwise stated.

[0179] As used herein, the following directional terms “frame facing side”, “non-frame facing side”, “forward”, “rearward”, “front”, “rear”, “up”, “down”, “above”, “below”, “upward”, “downward”, “top”, “bottom”, “side”, “vertical”, “horizontal”, “perpendicular” and “transverse” as well as any other similar directional terms refer to those directions of a bicycle in an upright, riding position and equipped with the bicycle component. Accordingly, these directional terms, as utilized to describe the bicycle component should be interpreted relative to a bicycle in an upright riding position on a horizontal surface and that is equipped with the bicycle component. The terms “left” and “right” are used to indicate the “right” when referencing from the right side as viewed from the rear of the bicycle, and the “left” when referencing from the left side as viewed from the rear of the bicycle.

[0180] The phrase “at least one of” as used in this disclosure means “one or more” of a desired choice. For one example, the phrase “at least one of” as used in this disclosure means “only one single choice” or “both of two choices” if the number of its choices is two. For another example, the phrase “at least one of” as used in this disclosure means “only one single choice” or “any combination of equal to or more than two choices” if the number of its choices is equal to or more than three. Also, the term “and/or” as used in this disclosure means “either one or both of”. For instance, the phrase “at least one of A and B” encompasses (1) A alone, (2), B alone, and (3) both A and B. The phrase “at least one of A, B, and C” encompasses (1) A alone, (2), B alone, (3) C alone, (4) both A and B, (5) both B and C, (6) both A and C, and (7) all A, B, and C. In other words, the phrase “at least one of A and B” does not mean “at least one of A and at least one of B” in this disclosure.

[0181] Also, it will be understood that although the terms “first” and “second” may be used herein to describe various components, these components should not be limited by these terms. These terms are only used to distinguish one component from another. Thus, for example, a first component discussed above could be termed a second component and vice versa without departing from the teachings of the present invention.

[0182] The term “attached” or “attaching”, as used herein, encompasses configurations in which an element is directly secured to another element by affixing the element directly to the other element; configurations in which the element is indirectly secured to the other element by affixing the element to the intermediate member(s) which in turn are affixed to the other element; and configurations in which one element is integral with another element, i.e. one element is essentially part of the other element. This definition also applies to words of similar meaning, for example, “joined”, “connected”, “coupled”, “mounted”, “bonded”, “fixed” and their derivatives. Finally, terms of degree such as “substantially”, “about” and “approximately” as used herein mean an amount of deviation of the modified term such that the end result is not significantly changed.

[0183] While only selected embodiments have been chosen to illustrate the present invention, it will be apparent to those skilled in the art from this disclosure that various changes and modifications can be made herein without departing from the scope of the invention as defined in the appended claims. For example, unless specifically stated otherwise, the size, shape, location or orientation of the various components can be changed as needed and/or desired so long as the changes do not substantially affect their intended function. Unless specifically stated otherwise, components that are shown directly connected or contacting each other can have intermediate structures disposed between them so long as the changes do not substantially affect their intended function. The functions of one element can be performed by two, and vice versa unless specifically stated otherwise. The structures and functions of one embodiment can be adopted in another embodiment. It is not necessary for all advantages to be present in a particular embodiment at the same time. Every feature which is unique from the prior art, alone or in combination with other features, also should be considered a separate description of further inventions by the applicant, including the structural and/or functional concepts embodied by such feature(s). Thus, the foregoing descriptions of the embodiments according to the present invention are provided for

illustration only, and not for the purpose of limiting the invention as defined by the appended claims and their equivalents.

Claims

1. A bicycle component comprising: a rechargeable power source configured to be charged by electric power from an external power supply; and an electrical switch configured to be activated by only electric power from the rechargeable power source.
2. The bicycle component according to claim 1, further comprising a holding structure configured to detachably hold the rechargeable power source to the bicycle component.
3. A bicycle component comprising: a rechargeable power source configured to be charged by electric power from an external power source; an electrical switch configured to be activated by electric power from the rechargeable power source; and a holding structure configured to detachably hold the rechargeable power source to the bicycle component.
4. The bicycle component according to claim 2, wherein the holding structure includes an electrical terminal structure electrically coupling the rechargeable power source to the bicycle component.
5. The bicycle component according to claim 2, wherein the holding structure includes an accommodating part and a power source cover, the accommodating part is configured to accommodate the rechargeable power source, the accommodating part includes an opening for removing the rechargeable power source from the holding structure, and the power source cover is configured to at least partially cover the opening of the accommodating part.
6. The bicycle component according to claim 5, wherein the holding structure includes a cover fixing structure configured to fix the power source cover relative to the accommodating part.
7. The bicycle component according to claim 1, further comprising a base member configured to be mounted to a bicycle; and an operating member movably coupled relative to the base member.
8. The bicycle component according to claim 7, wherein the operating member is configured to operate an operated device mounted to the bicycle.
9. The bicycle component according to claim 7, wherein the operating member is configured to activate the electrical switch in response to an input to the operating member.
10. The bicycle component according to claim 7, further comprising an additional operating member movably coupled relative to the base member, the additional operating member being different from the operating member.
11. The bicycle component according to claim 10, wherein the additional operating member is configured to operate an additional operated device mounted to the bicycle.
12. The bicycle component according to claim 10, wherein the base member includes a proximal end configured to be mounted to the bicycle, and a distal end opposite to the proximal end, the additional operating member is pivotally supported by an axle to pivot with respect to the base member, and the axle is disposed closer to the distal end than to the proximal end.
13. The bicycle component according to claim 10, wherein the electrical switch is provided to at least one of the operating member and the additional operating member.
14. The bicycle component according to claim 10, further comprising a hydraulic unit configured to operate the additional operated device in response to an input to the additional operating member.
15. The bicycle component according to claim 10, wherein the operating member is movably mounted to the additional operating member.
16. The bicycle component according to 1, further comprising a charging port configured to supply electrical power to the rechargeable power source from the external power supply.
17. The bicycle component according to claim 16, further comprising a cover member configured to cover the charging port in a case where the cover member is in a covering state.
18. The bicycle component according to claim 1, further comprising a wireless communicator electrically coupled to the rechargeable power source.

- 19.** The bicycle component according to claim 18, wherein the wireless communicator is configured to transmit information related to the rechargeable power source to an external device.
- 20.** The bicycle component according to claim 18, further comprising a substrate supporting the wireless communicator; and an antenna provided to the substrate and electrically coupled to the wireless communicator, the substrate being disposed between the rechargeable power source and the antenna.
- 21.** The bicycle component according to claim 20, wherein the antenna is disposed along a peripheral edge of the substrate.
- 22.** The bicycle component according to claim 5, further comprising a wireless communicator electrically coupled to the rechargeable power source; a substrate supporting the wireless communicator on a first side; and an antenna provided to a second side of the substrate and electrically coupled to the wireless communicator, the second side faces in an opposite direction to the first side, the rechargeable power source is closer to the power source cover than the antenna.
- 23.** The bicycle component according to claim 20, further comprising an electronic controller configured to control supply of electrical power from the rechargeable power source to the wireless communicator based on an output from a detector, the detector being configured to detect a state of at least one of the bicycle and the bicycle component.
- 24.** The bicycle component according to claim 1, further comprising an electronic controller configured to obtain information related to the rechargeable power source.
- 25.** The bicycle component according to claim 24, further comprising a notification device electrically coupled to the electronic controller, the electronic controller being configured to control the notification device based on the information.
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